News October 2017

Watch Grand Designs NZ, TV3 7.30pm Tues 17 Oct - AES system installed in Black Pavillion House

Tomorrow Tues 17th October at 7.30pm on TV3, Grand Designs NZ will feature the Black Pavillion House near Motueka, Nelson with a raised bed AES wastewater system. The property is adjacent to an inlet where tidal water rises within 5m of the northern boundary on spring tides. Combined with a high water table this requires the AES pipes to be installed in a raised bed to provide 0.6m vertical separation to the water table. Pipework suspended under the building subfloor connects to the septic tank which is also partially raised allowing effluent to gravity flow to the AES bed.

The design is for a four bedroom house treating 1400 l/day. The photo above shows the AES bed prior to covering with topsoil and planting with coastal shrubs. The job was designed by Gary Stevens and installed by Fusion Plumbing. The homeowners are looking forward to the ongoing low-cost performance of their AES system.

When high groundwater is a design factor, a passive AES septic system can often be achieved by mounding up the ground around the septic tank and bed, and suspending drainage pipework to conserve elevation.

If you miss it you can watch it on Demand at ThreeNow.

Latest Interim Test Results for AES system in OSET-NTP Trial 12

Environment Technology installed an AES system at the Rotorua Onsite Effluent Treatment, National Treatment Programme (OSET-NTP) testing facility in September 2016.

All the OSET-NTP results mentioned below are interim, unverified results and are to be regarded as such until the official report is released early next year. These interim results apply to the system tested in Trial 12 which ran from October 2016 – July 2017, which received 1000L/d. The AES treatment bed tested had a waterproof layer in the top 150mm of the bed to prevent rainwater accessing the bed and from affecting the treated effluent sample, and was lined on the sides and bottom to allow collection of the treated effluent. Normally an AES-38 would be installed below ground with no linings at all, but the system tested was not, due to site constraints, equivalency with the other tested systems which are all above ground, and as equivalent sample collection cannot take place with a bottomless treatment bed.

The test results have mirrored the advanced secondary results achieved in North America and Europe. The interim results from the benchmarking period are shown in the table below. We are retaining the system for Trial 13, where we will recirculate the treated effluent back through the anaerobic septic tank to further reduce total nitrogen (TN).

- A+ rating for BOD5
- A+ for TSS
- D rating for Total Nitrogen
- A rating for Ammoniacal nitrogen
- B rating for Total Phosphorus
- B rating for Faecal Coliforms

Test	Median	Std Deviation
BOD (g/m^3)	2	1.45
TSS (g/m^3)	3	13.0
TN (g/m^3)	36.98	5.0
NH_4^- (g/m^3)	1.67	0.5
TP (g/m^3)	2.59	0.4
FC (cfu/100mL)	5,730	6,624.3

Table 1. Interim results for the benchmarking period of the AES-38 system in Trial 12

A+ treatment quality for BOD and TSS was also achieved after five days of the system operating at 200% of design loading, and an A+ rating for energy use as the only power required was as a consequence of having to pump to the effluent sampling chamber at the OSET-NTP facility.

Rated Indicators for Median Value	Rating Letters and Corresponding Levels				
	A+	A	B	C	D
BOD Biological Oxygen Demand (g/m^3)	<5	<10	<20	<30	≥30
TSS Total Suspended Solids TSS (g/m^3)	<5	<10	<20	<30	≥30
TN Total Nitrogen (g/m^3)	<5	<15	<25	<30	≥30
NH_4^- Ammoniacal nitrogen (g/m^3)	<1	<5	<10	<20	≥20
TP Total phosphorous (g/m^3)	<1	<2	<5	<7	≥7
FC Faecal Coliforms (cfu/100mL)	<10	<200	<10,000	<100,000	≥100,000
Energy (KWh/d) mean	0	<1	<2	<5	≥5

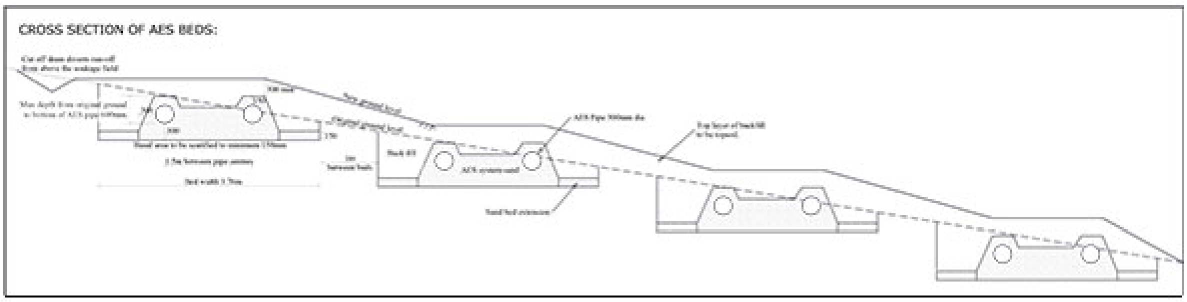
Table 2. OSET-NTP benchmark rating indicators

For more information check out OSET testing or call us on 0800 927 834 (0800 waste H2O).

Large scale AES installation at Hillsborough Hideaway in Taranaki

A larger scale AES system of 80 AES pipes was installed in July 2017, on a sloping site with Category 4 soils outside New Plymouth. The system uses a distribution box to evenly distribute the flow from the septic tank into four separate beds or sub-systems, each of 20 AES pipes. Laid on 4 descending levels, the four sub-systems are installed in parallel down the slope of the field, as shown in the drawing below. Each sub-system is made up of two AES pipe rows of 30 lineal metres, connected in series. This system layout is called a combination system.

The system services a complex of accommodation units, restaurant, outdoor activities and a new Holden car museum. It was designed for a maximum flow of 9195 litres per day based on 220 restaurant diners, 100 visitors per day, and 11 people residing in the house & homestay. Most of the facilities are on a hilltop plateau, from where the site slopes steeply to the west before levelling out prior to a wetland/watercourse. Kama Burwell of Greenbridge Sustainable Properties designed the system. The system was installed by Jason Baty of Vepo Plumbing, Stratford.



For more information about AES wastewater treatment systems visit the resources page on our website, watch the introductory videos, or call us on 0800 927 834 (0800 waste H2O).

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